

Activity 2

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1) push(54); push(52); pop(); push(55); push(62); s = pop()
enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue
(32); q = dequeue(); the value of s + q is —

Stack

push(54) $\rightarrow$ (54)

push(52) $\rightarrow$ (54, 52)

pop() = 52

push(55) $\rightarrow$ (54, 55)

push(62) $\rightarrow$ (54, 55, 62)

s = pop() = 62

so s = 62

Queue

enqueue(21) $\rightarrow$ (21)

enqueue(24) $\rightarrow$ (21, 24)

dequeue() $\rightarrow$ removes 21

enqueue(28) $\rightarrow$ (24, 28)

enqueue(32) $\rightarrow$ (24, 28, 32)

q = dequeue() = 24

q = 24

s + q = 62 + 24

= 86

$\therefore \boxed{s + q = 86}$

② postfix expression

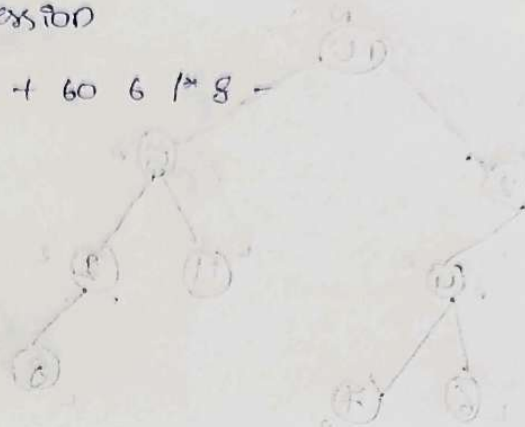
Expression $10\ 5\ +\ 60\ 6\ /\ * 8\ -$

$10 + 5 = 15$

$60 / 6 = 10$

$15 \times 10 = 150$

$150 - 8 = 142$



Ans = 142

③ Gate CSE

minimum number of enqueue

operations on $a_1 = 6$

Answer: 6

④ REVERSE operation
stack order O(n)

5) Postfix Expression

$8\ 2\ 3\ /\ 2\ 3\ * + 5\ /\ *$

After $3\ * 2$ is evaluated the top two stack element are 6, 5

⑥ Infix to postfix